CHEMISTRY

SECTION – I

Single Correct Choice Type

1. **(b)** $I_2 + 2Na_2S_2O_3 \longrightarrow 2NaI + Na_2S_4O_6$

Millimoles of $Na_2S_2O_3$ consumed = 6.

Millimoles of I_2 consumed = 3.

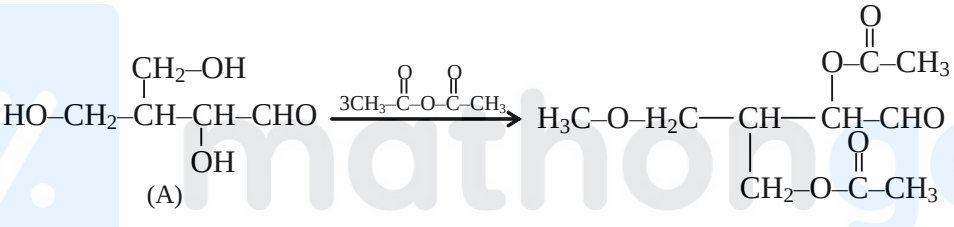
$3I_2 + 6NaOH \longrightarrow 5NaI + NaIO_3 + 3H_2O$

Millimole of I_2 reacted with NaOH = $\frac{30-6}{2} = 12$

Total millimole of I_2 consumed in this reaction = 15.

Initial concentration of I_2 = $\frac{15}{150}$ = **0.1 M**.

2. **(a)** Compound (A) is a metal nitride and (B) is NH_3 which gives white fumes of NH_4Cl with HCl. The metals which directly react with N_2 present in air on heating are Li, Mg and Al.

3. **(b)** 

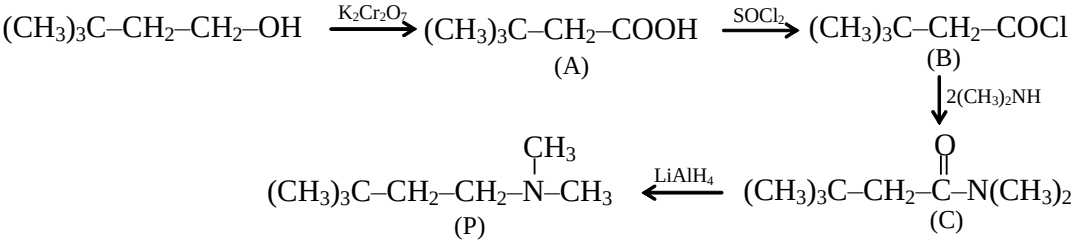
One mol of this compound can react with only 1 mol of HIO_4. It is oxidised by Br_2/H_2O to give carboxylic acid which forms a lactone ring of five member on heating.

4. **(d)** $6HCHO \rightleftharpoons C_6H_{12}O_6$

$c(1-\alpha)$ $\frac{c\alpha}{6}$

$$\frac{M_{theoretical}}{M_{observed}} = \frac{30}{150} = \frac{c(1-\alpha)+\frac{c\alpha}{6}}{c}$$

α = **0.96**.

5. **(b)** 

6. **(a)** $AlCl_3 \rightleftharpoons Al^{3+} + 3Cl^-$

$1-0.8 \qquad 0.8 \qquad 3\times0.8$

$$i = \frac{(3\times0.8)+0.8+(1-0.8)}{1} = 3.4$$

$$\underset{1-0.9}{BaCl_2} \rightleftharpoons \underset{0.9}{Ba^{2+}} + \underset{2\times0.9}{2Cl^-}$$

Proceeding in the same way $i = 3.7$ for Na_3PO_4 and $i = 3.8$ for $K_4[Fe(CN)_6]$.

SECTION – II
Multiple Correct Choice Type

7. **(a, b, c)**

At the point of maximum value of radius distribution function $\frac{dp(r)}{dr} = 0$.

$$\therefore \; 2r - \frac{2r^2z}{a_o} = 0$$

$$r = \frac{a_o}{z} \text{ pm.}$$

Radius of $Li^{2+} = \frac{a_o}{3}$ pm.

Radius of $He^+ = \frac{a_o}{2}$ pm.

Radius of H^+ ion is the radius of nucleus not the radius of shell.

8. **(a, b)** $3Cl_2O_6 + 6NaOH \longrightarrow 5NaClO_4 + NaOCl + 3H_2O$

9. **(a, b)** Due to rearrangement (neighbouring group participation by phenyl ring).

10. **(b, d)** $Cu_2S + 2Cu_2O \longrightarrow 6Cu + SO_2$

$HgS + 2HgO \longrightarrow 3Hg + SO_2$

$PbS + 2PbO \longrightarrow 3Pb + SO_2$

11. **(b, c, d)**

Structure (II) is non-superposable mirror image of (I). Structure (II) can be obtained with similar configuration when the molecule (III) is rotated out of plane by 180°. Therefore (II) and (III) are identical.

SECTION – III
Reasoning Type

12. **(d)** An optically active molecule may either be asymmetric or dissymmetric.

13. **(a)** For ideal gas Joule Thomson coefficient has zero value. Intermolecular interaction can be of attractive or repulsive nature.

14. **(d)** Any reaction can be under standard or non-standard conditions. If the standard conditions are used, statement-1 is correct or if non-standard conditions are employed, it may be incorrect.

15. **(b)**

16. **(c)** Due to steric inhibition of resonance, the basicity of N-N-dimethylpicramide is more than picramide. +I effect is just a supporting factor but the main factor is steric factor.

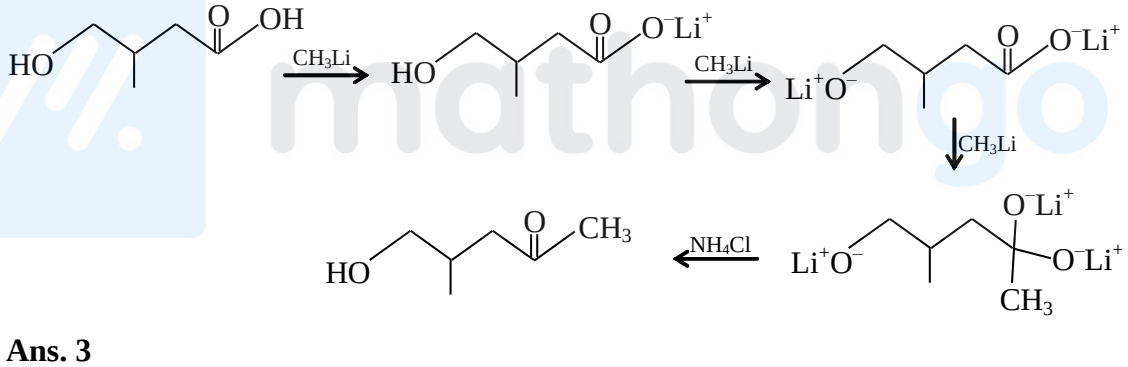
SECTION–IV
Matrix–Match Type

1. **(A) – 1, 5 ; (B) – 4 ; (C) – 3 ; (D) – 2, 5**

2. **(A) – 2, 4 ; (B) – 1, 2, 3, 4, 5 ; (C) – 2, 3, 5 ; (D) – 1, 2, 3**

SECTION–V
Subjective or Numerical Type

1.



Ans. 3

2. $Z = 1 + \frac{Pb}{RT}$ (at high pressure)

$$\text{Slope} = \frac{b}{RT} = \frac{\pi}{492.6}$$

$$b = \frac{\pi}{492.6} \times 0.0821 \times 300$$

$$\text{and } b = \frac{4}{3}\pi r^3 \times 4N_A$$

$$= \frac{\pi}{492.6} \times 0.0821 \times 300 \times 10^{-3}$$

$r = 2.5$ Å ; $d = 5$ Å

Ans. 5

3. Let initial activities of X and Y be x_o and y_o. After 2 half-lives the activity of x will be $\frac{x_o}{4}$

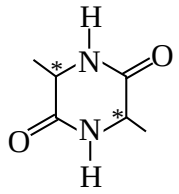
$x_o + y_o = 8000$; $\frac{3x_o}{4} = 4500$

$x_o = 6000$; $y_o = 2000$

$\frac{x_o}{y_o} = 3.$

Ans. 3

4.



This is symmetrical and has 2-chiral centres so

$2^{n-1} + 2^{\frac{n-2}{2}} = 3$ isomers.

Ans. 3

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